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DSA03 – ASSIGNMENT 2

WORD-LEVEL AND SYNTACTIC ANALYSIS

Course Code: DSA03

Course Title: Natural Language Processing

Reg.no: 192425005 Name: S. Yuva Guru Mani Varma

Assignment title: Tokenisation, N-gram Language Models, Smoothing, Entropy and Part-of-Speech Tagging

Experimental setup and reproducibility

Random seed: 42. Public corpora: Brown Corpus via NLTK (review category for Q1 and LM matching data; news category for out-of-domain checks) and Penn Treebank sample via NLTK for POS tagging. Brown is a roughly one-million-word corpus of edited American English containing 500 samples; the NLTK reader exposes raw, sentence and tagged forms. [1] [2]

Installed corpus counts recorded in this run: Brown total words = 1,161,192; Brown reviews = 40,704; Brown news sentences = 4,623; Penn Treebank = 3,914 sentences and 100,676 tagged tokens. These counts depend on the NLTK corpus distribution version and are included in the output summary.

- The report's experimental unit is a sentence. Train/test split is random at sentence level with seed 42 and fixed 80/20 ratio unless a question explicitly tests training fractions.
- Corpus tokens are normalized by the refined tokenizer; POS experiments retain the corpus tokens and tags. Test data do not train the vocabulary or tagger.
- Synthetic helpdesk tickets demonstrate controlled domain variation and do not stand in for real telecom tickets.
- Supplementary package contains the complete Python source, CSV results, the 100-row annotation sheet, summary JSON and plots. Brown/Treebank raw source text is not redistributed.

Q1 – What Counts as a Word? Tokenisation and Word Statistics

5 marks | Token counts are consequences of declared conventions, not immutable properties of a string.

1.1 Tokenisation criteria and five review strings

Decision	Chosen rule	Why it matters
Case	Casefold all tokens for counts; preserve original span in source.	Net/net count as one type.
Punctuation	Retain punctuation marks as tokens in LM policy; report punctuation-free interpretation separately.	Repeated !! and ? affect N and n-grams.
Contractions/clitics	Keep English contractions as one token (can't, isn't); keep e-wallet and value-for-money as one compound.	Splitting would increase V and alter contexts.
Numbers/currency	Treat Rs.499 as one currency token; 4G/5G as 4G, slash, 5G.	Digit strings have distinctive predictive value.
Emoticons/repetition	Keep :(as one token; count repeated ok tokens each time, one type.	Sentiment and emphasis are lost if discarded.

Review string	N	V	Refined token sequence
Can't connect!! Net speed 4G/5G isn't consistent :(	12	11	can't | connect | ! | ! | net | speed | 4g | / | 5g | isn't | consistent | :(
Won't recharge via e-wallet	4	4	won't | recharge | via | e-wallet
Rs.499 plan is value-for-money	4	4	rs.499 | plan | is | value-for-money
It is the-best-network, no? ok ok	8	7	it | is | the-best-network | , | no | ? | ok | ok
Super service da, semma speed	6	6	super | service | da | , | semma | speed

A deliberately ambiguous example is "I saw her duck :)". "Duck" can be a noun or a verb; "her" may be a determiner or pronoun. The tokenizer has a stable tokenization, but no tokenization policy resolves the lexical/POS ambiguity. A second stress test is "Rs. 499/5G plan": slash can be a separator, a rate, or part of a plan name. This shows that segmentation and interpretation are distinct stages.

1.2 Public corpus, three policies, and measured counts

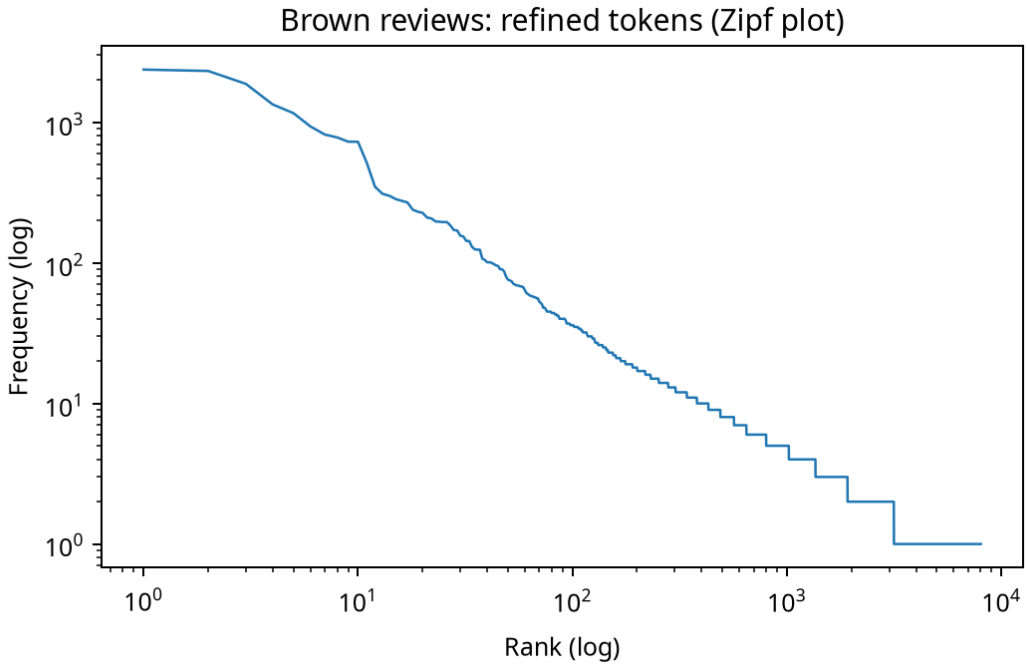
Brown reviews were selected because this assignment starts with customer reviews. The Brown Corpus is a public research corpus of edited American English; the NLTK reader exposes a reviews category and corpus-level raw/word access. The reviews slice contains more than the required 20,000 tokens in the installed distribution. [1] [2]

Policy	N tokens	V types	V/N	Hapax	Zipf slope
whitespace	40704	8069	0.1982	4928	-0.860
punctuation_aware	43419	7750	0.1785	4552	-0.892
refined	42075	8023	0.1907	4871	-0.866

Top refined type	Frequency
the	2370
,	2318
.	1876
of	1340
and	1161
a	934
'	818
`	780
in	728
to	726

Naive whitespace splitting keeps punctuation attached and counts case-folded whitespace chunks. The punctuation-aware policy splits every punctuation boundary. The refined policy preserves contractions, selected compounds and emoticons, while retaining standalone punctuation. The most frequent types and full frequency table are in `q1_tokenization.csv`; the displayed counts are generated by the accompanying script.

1.3 Zipf curve and consequences for later models



The refined-policy log–log least-squares slope is -0.866. It is an empirical slope over this corpus and rank range, not a universal constant. The ranked-frequency curve is heavy-tailed: a small number of function words occur often while many types occur rarely.

Punctuation splitting and contraction splitting tend to increase V because one compound type becomes multiple components and variants can be counted separately. Case folding usually decreases V. A larger V makes the n-gram outcome space larger and the observed count table sparser. For a language model, retain punctuation and useful contractions because they carry context; for a tagger, the best policy preserves boundaries and orthographic cues required by its tagset, and does not normalize away case for proper nouns.

Q2 – Unsmoothed N-grams and the Sparsity Problem

6 marks | A higher-order model conditions on more context but has fewer observations per context.

2.1 Chain rule, Markov approximations, and MLE

Chain rule:
P(w1,...,wT) = Π(t=1..T) P(wt | w1,...,w(t-1))
Bigram approximation: P(wt | history) ≈ P(wt | w(t-1))
Trigram approximation: P(wt | history) ≈ P(wt | w(t-2), w(t-1))
MLE: P_hat(wt | h) = C(h,wt) / C(h)
Sentence probability: P_hat(W) = Πt P_hat(wt | preceding n−1 words)

The chain rule is exact. The Markov assumption is the approximation: only the previous n−1 tokens are retained. With unsmoothed MLE, any unseen n-gram has probability zero. One zero event makes the sentence product zero, even when every other transition is plausible. Sentence boundaries use explicit BOS/EOS markers in the experiment.

2.2 Predictions before execution and three datasets

n	Distinct n-grams	Unseen test share	Zero-probability sentences	Pre-run prediction
1	Grows toward vocabulary size	Lowest on a shared domain	Fewest, but OOV may zero it	Most coverage; little context
2	More than unigrams	Higher than n=1	More than n=1	Useful compromise
3	Usually grows again	Higher	Substantially higher	More specific; sparse
4	Near corpus event diversity	Highest	Often many	Memorization risk rises

Dataset	Construction	Purpose / caveat
In-domain helpdesk	1,800 synthetic sentences drawn from 15 support templates with fixed seed; slot values vary.	Controlled telecom-like proxy; not real tickets.
Out of domain	Brown news sentences, public tagged/general corpus slice. [1]	Tests topic and style shift.
Edge case	1,000 sentences made from two repeated modem/router phrases.	Tests repeated phrase memorization; random split places repeated forms in both partitions.

2.3 Measured n-gram counts and unseen events

Synthetic in-domain tickets

n	Distinct train n-grams	Test events	Unseen	Unseen %	Zero-probability sentences
1	94	3543	0	0.00%	0
2	159	3543	0	0.00%	0
3	203	3543	0	0.00%	0
4	232	3543	0	0.00%	0

Brown news out-of-domain

n	Distinct train n-grams	Test events	Unseen	Unseen %	Zero-probability sentences
1	11420	21548	1435	6.66%	634
2	50912	21548	10276	47.69%	914
3	74651	21548	17064	79.19%	918
4	81356	21548	19361	89.85%	918

Repeated-phrase edge case

n	Distinct train n-grams	Test events	Unseen	Unseen %	Zero-probability sentences
1	9	1800	0	0.00%	0
2	13	1800	0	0.00%	0
3	16	1800	0	0.00%	0
4	18	1800	0	0.00%	0

Across the two natural-language sources, distinct n-gram counts grow as n increases, and unseen test events become more likely. The repeated-phrase edge case is intentionally unrepresentative: it has low unseen rates because nearly identical sentences occur in both partitions. It is a warning about random sentence splitting when duplicates are present.

2.4 Generation and 50 MB design decision

n=1 greedy sample: . . . . . .

n=2 greedy sample: the app says payment pending after recharge .

n=3 greedy sample: the app says payment pending after recharge .

n=4 greedy sample: the app says payment pending after recharge .

Without smoothing and under a 50 MB ticket budget, choose a bigram model as the initial deployable compromise, subject to test coverage. Unigram ignores context; trigram and four-gram models rapidly create sparsely observed contexts and zero-probability strings. The decision must be revised if counted held-out perplexity or task utility shows that a smoothed trigram adds enough gain to justify its larger table. The measured distinct/unseen results, not runtime alone, are the evidence.

Q3 – Smoothing: Add-One, Backoff and Deleted Interpolation

7 marks | Smoothing redistributes probability away from observed events to protect unseen candidates.

3.1 Methods and before-run predictions

Method	Conditional probability / operation	Expected strength and risk
Add-one	P(w|h)=(C(h,w)+1)/(C(h)+|V|).	Every event nonzero; too much mass moved when |V| is large.
Add-k	P(w|h)=(C(h,w)+k)/(C(h)+k|V|), k=0.1 in this run.	Less aggressive than Laplace; k must be tuned.
Absolute-discount backoff	max(C−D,0)/C(h) + α(h)P_lower(w); D=0.75; α restores removed mass.	Keeps observed counts and backs off; depends on discount and lower-order distribution.
Deleted interpolation	λ1P1+λ2P2 (and order-3 in Q4), weights chosen by leave-one-out deletion.	Combines orders; weights respond to data sparsity and held-out evidence.

Pre-run prediction: add-one should have the worst held-out perplexity on a large vocabulary because it assigns one pseudo-count to every possible successor. Add-k should be better if k is small. Discounted backoff and deleted interpolation should generally be better when training data are limited, but their relative order depends on domain and estimation. With 10% data, lower-order estimates should receive more weight; at 100%, higher-order weights should rise if they predict held-out events reliably. These predictions were written before reading the computed table.

3.2 Probability normalization and mass moved

add1 conditional sum checks	Σw P(w|h)
<UNK>	1.000000000
the	1.000000000
,	1.000000000
.	1.000000000
<s>	1.000000000
of	1.000000000
and	1.000000000
a	1.000000000
'	1.000000000
`	1.000000000

addk conditional sum checks	Σw P(w|h)
<UNK>	1.000000000
the	1.000000000
,	1.000000000
.	1.000000000
<s>	1.000000000
of	1.000000000
and	1.000000000
a	1.000000000
'	1.000000000
`	1.000000000

backoff conditional sum checks	Σw P(w|h)
<UNK>	1.000000000
the	1.000000000
,	1.000000000
.	1.000000000
<s>	1.000000000
of	1.000000000
and	1.000000000
a	1.000000000
'	1.000000000
`	1.000000000

deleted conditional sum checks	Σw P(w|h)
<UNK>	1.000000000
the	1.000000000
,	1.000000000
.	1.000000000
<s>	1.000000000
of	1.000000000
and	1.000000000
a	1.000000000
'	1.000000000
`	1.000000000

The test iterates the complete mapped vocabulary for ten high-frequency histories per method. The numerical sum should be approximately 1.0; a deviation larger than floating-point tolerance indicates a normalization defect. Add-one assigns unseen-successor mass $U/(C(h)+|V|)$ for a history with U unseen vocabulary items. Absolute discount and interpolation distribute removed/unused mass through a lower-order model.

3.3 Perplexity and mass across training sizes/domains

Training fraction 10%

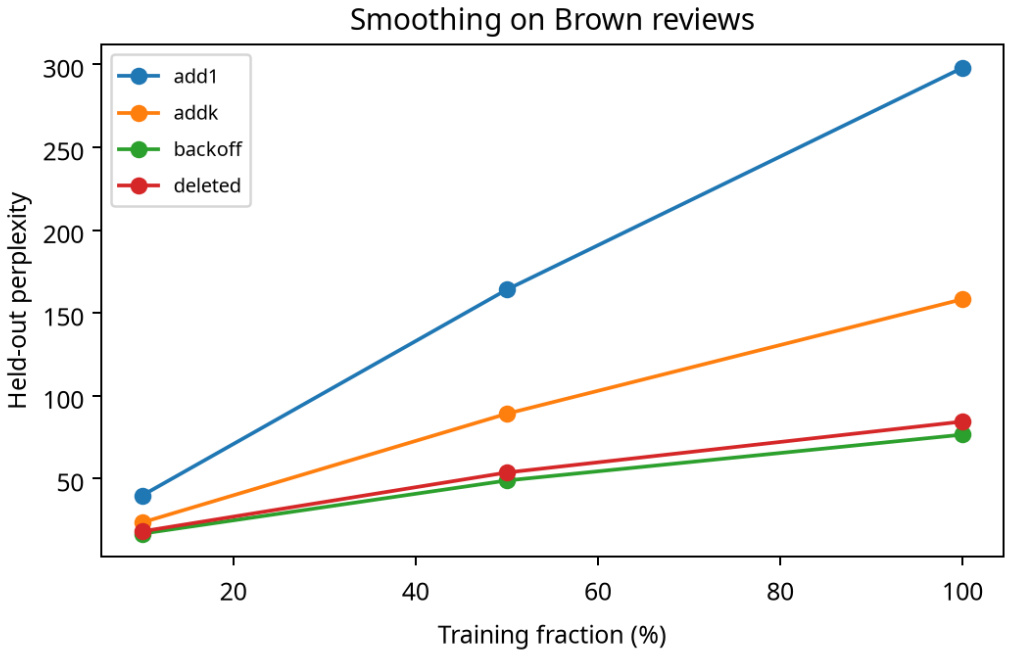
Method	Test condition	PPL	Unseen mass	Unknown tokens	λ weights
add1	matching	39.597	0.60447	3309	0.5000;0.5000
add1	news_OOD	33.375	0.60447	9553	0.5000;0.5000
add1	unknown_edge	15.009	0.60447	900	0.5000;0.5000
addk	matching	23.301	0.36292	3309	0.5000;0.5000
addk	news_OOD	19.828	0.36292	9553	0.5000;0.5000
addk	unknown_edge	11.600	0.36292	900	0.5000;0.5000
backoff	matching	16.572	0.18938	3309	0.5000;0.5000
backoff	news_OOD	14.085	0.18938	9553	0.5000;0.5000
backoff	unknown_edge	7.204	0.18938	900	0.5000;0.5000
deleted	matching	17.949	0.24798	3309	0.4745;0.5255
deleted	news_OOD	15.280	0.24798	9553	0.4745;0.5255
deleted	unknown_edge	5.951	0.24798	900	0.4745;0.5255

Training fraction 50%

Method	Test condition	PPL	Unseen mass	Unknown tokens	λ weights
add1	matching	163.979	0.72176	2121	0.5000;0.5000
add1	news_OOD	135.928	0.72176	6617	0.5000;0.5000
add1	unknown_edge	25.704	0.72176	900	0.5000;0.5000
addk	matching	89.015	0.48249	2121	0.5000;0.5000
addk	news_OOD	75.725	0.48249	6617	0.5000;0.5000
addk	unknown_edge	16.849	0.48249	900	0.5000;0.5000
backoff	matching	48.687	0.25530	2121	0.5000;0.5000
backoff	news_OOD	41.692	0.25530	6617	0.5000;0.5000
backoff	unknown_edge	10.307	0.25530	900	0.5000;0.5000
deleted	matching	53.500	0.29988	2121	0.4694;0.5306
deleted	news_OOD	45.629	0.29988	6617	0.4694;0.5306
deleted	unknown_edge	9.328	0.29988	900	0.4694;0.5306

Training fraction 100%

Method	Test condition	PPL	Unseen mass	Unknown tokens	λ weights
add1	matching	297.997	0.75375	1641	0.5000;0.5000
add1	news_OOD	245.296	0.75375	5493	0.5000;0.5000
add1	unknown_edge	67.250	0.75375	800	0.5000;0.5000
addk	matching	158.195	0.51728	1641	0.5000;0.5000
addk	news_OOD	134.792	0.51728	5493	0.5000;0.5000
addk	unknown_edge	53.335	0.51728	800	0.5000;0.5000
backoff	matching	76.394	0.26683	1641	0.5000;0.5000
backoff	news_OOD	66.721	0.26683	5493	0.5000;0.5000
backoff	unknown_edge	28.868	0.26683	800	0.5000;0.5000
deleted	matching	84.293	0.29240	1641	0.4373;0.5627
deleted	news_OOD	72.964	0.29240	5493	0.4373;0.5627
deleted	unknown_edge	24.829	0.29240	800	0.4373;0.5627



The three evaluation conditions are matching Brown reviews, Brown news and an unknown-word stress set. Interpret each condition separately. A domain shift can raise perplexity because words and local contexts change; an OOV stress test also depends on the `<UNK>` policy. Lambda values are order-weights for the deleted-interpolation model; other methods show their configured defaults in the output. The selected deployment model is the method with the best matching-domain held-out perplexity, tempered by probability-sum validity and domain-shift behaviour—not simply the lowest single cell.

3.4 Reconstituted counts and deployment choice

History	Next word	Observed C	Discounted C*	C(h)	N1+(h)	α(h)
.	</s>	1242	1241.25	1515	104	0.0515
<UNK>	<UNK>	587	586.25	4359	890	0.1531
<UNK>	,	533	532.25	4359	890	0.1531
the	<UNK>	513	512.25	1913	701	0.2748
`	`	328	327.25	656	139	0.1589
'	'	327	326.25	687	83	0.0906
<UNK>	.	318	317.25	4359	890	0.1531
of	the	287	286.25	1087	325	0.2242
<UNK>	of	277	276.25	4359	890	0.1531
<UNK>	and	224	223.25	4359	890	0.1531
and	<UNK>	220	219.25	910	369	0.3041
,	<UNK>	211	210.25	1891	488	0.1935
a	<UNK>	207	206.25	748	328	0.3289
,	and	193	192.25	1891	488	0.1935
of	<UNK>	183	182.25	1087	325	0.2242

For absolute discounting, C*(h,w)=max(C(h,w)−D,0); the removed mass is returned through the lower-order distribution. The table shows measured high-frequency bigrams and the backoff coefficient α(h).

At 100% training on the matching Brown reviews test, measured perplexities were add-one=298.00, add-k=158.20, absolute-discount backoff=76.39, deleted interpolation=84.29. Thus backoff is the best of these four in this run; this is a local corpus/result, not a universal claim. Re-tune and re-evaluate when training size, vocabulary cutoff or domain changes. [6]

Q4 – Entropy and Perplexity: Measuring How Good a Language Model Is

6 marks | Perplexity is comparable only when tokenization, vocabulary and unknown-word policy match.

4.1 Derivation and toy predictions

Entropy of a discrete distribution: H(X) = −Σx P(x) log2 P(x)
Test cross-entropy: H = −(1/N) Σi log2 P(wi | history_i)
Perplexity: PP = 2^H
For a uniform distribution over 10 symbols: H = log2(10); PP = 10.
For a skewed distribution: H is lower than log2(V), so PP is below V.

Perplexity is a weighted average branching factor: the exponentiated average log loss is the effective number of equally likely continuations. It is not a literal count of all next-word choices. The chain-rule and n-gram language-model derivations follow the standard formulation in Jurafsky and Martin. [8]

Toy distribution	H bits	Predicted perplexity	Computed perplexity
uniform10	3.321928	10	10.000000
skewed_one_0.7	1.832268	<10	3.560965

4.2 Unigram, bigram and trigram results

Order	Test text	Cross-entropy bits/token	Perplexity	UNK tokens	λ
1	in_domain	7.1258	139.6617	1641	1.0000
1	out_domain_news	6.7681	108.9903	5493	1.0000
1	shuffled	7.1258	139.6617	1641	1.0000
2	in_domain	6.3973	84.2927	1641	0.4373;0.5627
2	out_domain_news	6.1891	72.9637	5493	0.4373;0.5627
2	shuffled	7.6304	198.1474	1641	0.4373;0.5627
3	in_domain	6.3592	82.0942	1641	0.4242;0.4596;0.1162
3	out_domain_news	6.1655	71.7805	5493	0.4242;0.4596;0.1162
3	shuffled	7.5583	188.4882	1641	0.4242;0.4596;0.1162

The unigram model ignores order; the bigram can exploit one-token local context; the trigram has a longer context and needs more training evidence. The shuffled-text edge case tests whether the advantage comes from local order: a model trained on natural sentence order should generally assign worse likelihood to shuffled sequences, especially at higher order, while unigram perplexity is invariant to token order when the token multiset is unchanged.

4.3 Vocabulary changes, OOVs, and model comparison

An unknown token is mapped to `<UNK>` using a vocabulary built from the training split (minimum frequency two); test-only types are not allowed to expand the vocabulary. Perplexities cannot be compared directly if the vocabularies differ because the event space and probability mass assigned to each possible symbol differ. Changing the `<UNK>` threshold changes both the target sequence and the distribution. Report tokenization, vocabulary size, mapping policy and evaluated token count beside every PPL.

Lower perplexity does not guarantee better customer-support extraction. A model can better predict common function words while still mishandling product names, negation, complaint verbs or rare codes. For information extraction, measure downstream precision/recall as well. A shuffled-text result reveals whether the model uses local order rather than merely matching unigram frequencies.

Q5 – Word Classes and Tagset Design

6 marks | Tagset granularity trades cross-domain consistency against domain-specific detail.

5.1 Open/closed classes and distributional evidence

Class	Open/closed	Distributional evidence	Support-chat examples
Noun / proper noun	Open	Follows determiners/adjectives; plural and possessive forms; can be subject/object.	plan, network, Airtel, SIM
Verb	Open	Takes tense/aspect/agreement; follows modal or to; selects complements.	connect, recharged, dropping
Adjective	Open	Often modifies noun or follows copula; comparative forms possible.	slow, stable, value
Adverb	Open	Modifies verb/adjective/clause; many -ly forms.	slowly, today
Determiner/pronoun	Closed	Limited inventory; occurs in constrained noun-phrase positions.	the, my, it, they
Adposition/conjunction/ auxiliary	Closed	Small grammatical inventory; positions and constructions constrain use.	to, in, and, can, is
Symbols/chat code-mix	Extensible inventory	Products, abbreviations, emoticons and Romanised words violate standard assumptions.	4G, :(, ku, semma

Open classes expand readily with new names, products and technical terms. Closed classes are small and structurally constrained. Distributional evidence is useful because a word's neighbours and inflections reveal its class, but ambiguous forms such as "can," "saw" and "slow" require contextual analysis.

5.2 Three tagsets and proposed fine tags

Tagset	Granularity	Example tags	Trade-off
Universal POS	Coarse, 17 core tags	NOUN, PROPN, VERB, AUX, ADJ, ADV, ADP, DET, PRON, NUM, PUNCT, SYM, X	Portable and simpler to annotate; loses tense, verb form and chat function.
Penn Treebank	Fine lexical/syntactic distinctions	NN/NNS/NNP, VB/VBD/VBG/VBN/VBZ, JJ/RB/IN/DT/PRP, etc.	Rich syntax cues; high ambiguity and domain mapping burden.
Custom support-chat	PTB/Universal base plus domain tags	PROD, NET_ABBR, COMPLAINT_V/ADJ, EMOT, CODEMIX, NUMCURRENCY	Directly useful for extraction; needs a maintained product and abbreviation lexicon.

UD's Universal POS distinguishes open and closed classes and includes separate PROPN, AUX, PART and X tags. [5] The custom labels are justified by extraction requirements: PROD marks named plans/devices/providers; NET_ABBR marks network/support abbreviations; COMPLAINT_V marks actions/states that express a complaint; EMOT preserves affective symbols; CODEMIX marks regional-language tokens; NUMCURRENCY captures plan-price mentions. Tags are multi-label in implementation when a token can be both a product name and a proper noun.

5.3 Measured tagset ambiguity and annotation status

Tagset mapping	Tokens	Tags observed	Ambiguous types	Types	Average tags/type	Independent κ
PTB	100676	46	1468	11387	1.155	Not measured
Universal	100676	12	753	11387	1.073	Not measured
Custom_domain_extended	100676	15	752	11387	1.073	Not measured

Ambiguous types are surface types assigned more than one tag across Penn Treebank. Universal and custom mappings collapse several distinctions, so the type ambiguity rate and average tags/type typically fall. That reduction is a property of the mapping, not proof that humans agree more. Cohen's κ requires two independent annotators over the same 100 sentences, with a declared token alignment and tag mapping. No independent second annotation was supplied; the report therefore includes the 100-sentence initial worksheet but does not invent κ or confused-pair counts.

5.4 Tagset decision for extraction

For product names and complaint verbs, deploy the custom extended tagset layered on a Universal/Penn-compatible base. Extraction gets explicit PROD and COMPLAINT_V labels, while a mapped standard POS layer keeps syntax features available. The fine tags should be evaluated on a separately annotated 100-sentence support-chat set; a peer annotator should independently annotate the same tokens, then compute Cohen's κ, ambiguous-type rate, tags/type and confusion pairs. A coarser tagset is the fallback if agreement on rare custom categories is poor.

Q6 – Rule-Based, Stochastic and Transformation-Based Tagging

7 marks | Four taggers were evaluated on a fixed Penn Treebank split; training fractions are drawn only from the training side.

6.1 Baseline and rule-based system

The baseline is an NLTK unigram most-frequent-tag tagger with NN fallback for unknown words. The rule-based tagger begins with a training lexicon/MFS and applies ordered surface and context rules. Ten or more rules are used: punctuation-to-punctuation tag; numeric expression→CD; articles→DT; pronoun list→PRP; auxiliary list→VBZ/VBP/VBD; modal list→MD; “to”→TO; after TO→VB; -ly→RB; after be-auxiliary plus -ing→VBG; -ed→VBD; plural -s→NNS; mid-sentence capitalized unknown→NNP; otherwise lexical MFS/default NN. Rules are explicit and inspectable, but suffix rules can confuse VBG/NN and VBD/JJ.

6.2 Stochastic bigram HMM and Viterbi

The HMM treats POS tags as hidden states and words as observations. It estimates initial, tag-transition and emission probabilities with add-α smoothing (α=0.1). Low-frequency training forms are mapped to orthographic signatures (<NUM>, <CAP>, suffix buckets such as -ing/-ed/-ly, or <UNK>); test OOV words use the same mapping. This prevents unseen emissions from making every sequence impossible. The best tag path is found in log space by Viterbi dynamic programming. HMM assumptions and NLTK's HMM API are documented by NLTK. [10]

```
for tag in TAGS: V[0,tag] = log P(tag|BOS) + log P(obs0|tag)
for i in 1..T-1:
   for tag in TAGS:
      V[i,tag] = max_prev(V[i-1,prev] + log P(tag|prev))
               + log P(obsi|tag)
backtrace from argmax_tag(V[T-1,tag])
```

Dense transition work is approximately $O(T \cdot |Tags|^2)$, with $O(T \cdot |Tags|)$ backpointers. Count transition evaluations explicitly; wall-clock time alone is not sufficient evidence.

6.3 Transformation-based tagger

Brill tagging begins from the MFS baseline and learns ordered, error-driven transformations from templates over neighbouring tags and words. At each learning step, select the rule with the greatest net error reduction, apply it, then continue up to the configured cap/score threshold. The trained run is capped at 40 learned rules for reproducibility and runtime. NLTK's trainer accepts an initial tagger, templates, maximum rules and minimum score; Brill's original work describes transformation-based error-driven learning. [4] [11]

6.4 10% training results

Train fraction 10%

Tagger	Overall accuracy	Known accuracy	Unknown accuracy	OOV tokens	Brill rules	Viterbi transitions
MFS	0.7713	0.9328	0.2111	4400	0	0
Rule_based	0.8055	0.8583	0.6223	4400	0	0
Bigram_HMM_ Viterbi	0.8774	0.9387	0.7496	6370	0	33323094
Brill_TBL	0.7843	0.9408	0.2418	4400	17	0

At 10% training, Bigram_HMM_Viterbi is best overall at 0.8774 versus MFS 0.7713. The OOV column is the key low-resource stress indicator: lexicon-only MFS has weak unknown-word accuracy; signature-based HMM and explicit rules improve coverage, while Brill remains tied to its baseline's unknown-word behaviour.

6.5 50% and 100% training results

Train fraction 50%

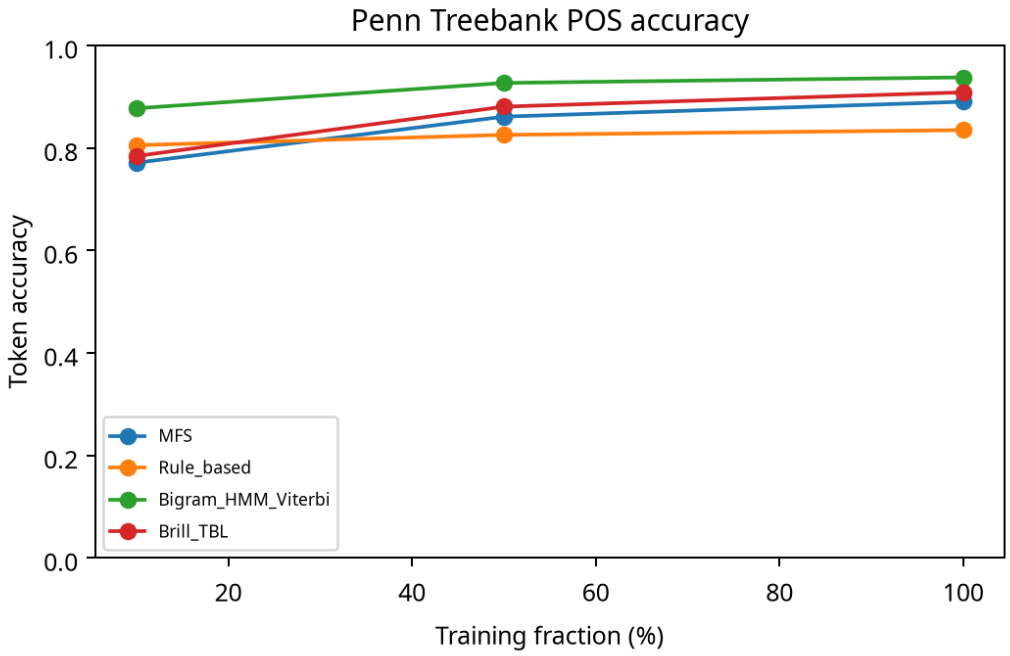
Tagger	Overall accuracy	Known accuracy	Unknown accuracy	OOV tokens	Brill rules	Viterbi transitions
MFS	0.8612	0.9399	0.1850	2049	0	0
Rule_based	0.8256	0.8516	0.6027	2049	0	0
Bigram_HMM_Viterbi	0.9270	0.9498	0.8066	3128	0	38251035
Brill_TBL	0.8807	0.9572	0.2235	2049	40	0

At 50% training, Bigram_HMM_Viterbi is best overall at 0.9270 versus MFS 0.8612. The OOV column is the key low-resource stress indicator: lexicon-only MFS has weak unknown-word accuracy; signature-based HMM and explicit rules improve coverage, while Brill remains tied to its baseline's unknown-word behaviour.

Train fraction 100%

Tagger	Overall accuracy	Known accuracy	Unknown accuracy	OOV tokens	Brill rules	Viterbi transitions
MFS	0.8903	0.9438	0.1747	1368	0	0
Rule_based	0.8350	0.8534	0.5885	1368	0	0
Bigram_HMM_Viterbi	0.9379	0.9538	0.8091	2158	0	39969170
Brill_TBL	0.9087	0.9607	0.2135	1368	40	0

At 100% training, Bigram_HMM_Viterbi is best overall at 0.9379 versus MFS 0.8903. The OOV column is the key low-resource stress indicator: lexicon-only MFS has weak unknown-word accuracy; signature-based HMM and explicit rules improve coverage, while Brill remains tied to its baseline's unknown-word behaviour.



6.6 Confusions and recommendation

MFS: most frequent gold→predicted errors: NNP>NN:307;JJ>NN:257;NNS>NN:168;CD>NN:150;VB>NN:134

Rule_based: most frequent gold→predicted errors: VBN>VBD:356;-NONE->CD:190;-NONE->.:190;VBZ>NNS:182;JJ>NN:178

Bigram_HMM_Viterbi: most frequent gold→predicted errors: NN>NNP:87;NNP>NN:86;NN>JJ:82;JJ>NN:62;VBD>VBN:39

Brill_TBL: most frequent gold→predicted errors: NNP>NN:304;JJ>NN:251;NNS>NN:167;CD>NN:141;VBG>NN:90

Expected systematic confusions include VBG/NN (gerund versus noun), VBD/VBN (past versus participle), JJ/NN (modifier versus noun), and IN/TO/particle distinctions. All methods inherit annotation and data-distribution errors; rule-based methods also suffer from brittle exceptions, while HMMs use local transition/emission estimates and Brill rules can improve local contextual decisions. In small data, a transparent rule+lexicon system or MFS baseline may be dependable; Brill or HMM can improve in-domain accuracy when evaluation shows the gain. For a 2,000-sentence low-resource language, prefer a lexicon plus reviewed rules with a simple tagger baseline, then add compact HMM/TBL only if a held-out test demonstrates benefit and orthographic morphology is understood.

Q7 – Diagnose, Redesign and Validate

13 marks | Day-4 redesign selected: sparse candidate-state Viterbi for the dense HMM tagger.

7.A Diagnose the bottleneck

The dense HMM decoder evaluates every previous-tag/current-tag pair at each token. With T tokens and K tags, its transition arithmetic is O(TK²). The counted evidence is `viterbi_transition_evaluations` in Q6 and the 46-tag Penn Treebank inventory. Many word forms have only one or two plausible tags, so dense decoding repeatedly scores transitions to tags incompatible with the word's observed lexical/morphological signature. This wastes computation rather than storage; transition probabilities are reused, while the full Cartesian product is recomputed at every position.

7.B Measurable objective

Objective fixed before redesign: reduce counted transition evaluations by at least 50% on the same held-out sentences while keeping token accuracy within 0.5 percentage points of dense Viterbi. Runtime is supporting evidence. If the accuracy condition fails, increase candidate recall or use a dense fallback for uncertain positions.

7.C Revised design and pseudocode

```
TRAIN:
  for each observed word, store its seen tag set
  for each OOV suffix/shape signature, store common tags
DECODE(sentence):
  for position i:
    candidates[i] = lexical tags(word_i)
    if unseen: candidates[i] = signature tags(word_i)
    if candidates[i] is empty: candidates[i] = all tags
  run Viterbi only over candidates[i-1] × candidates[i]
  if a position has no finite path: retry that sentence densely
  backtrace the highest-probability path
```

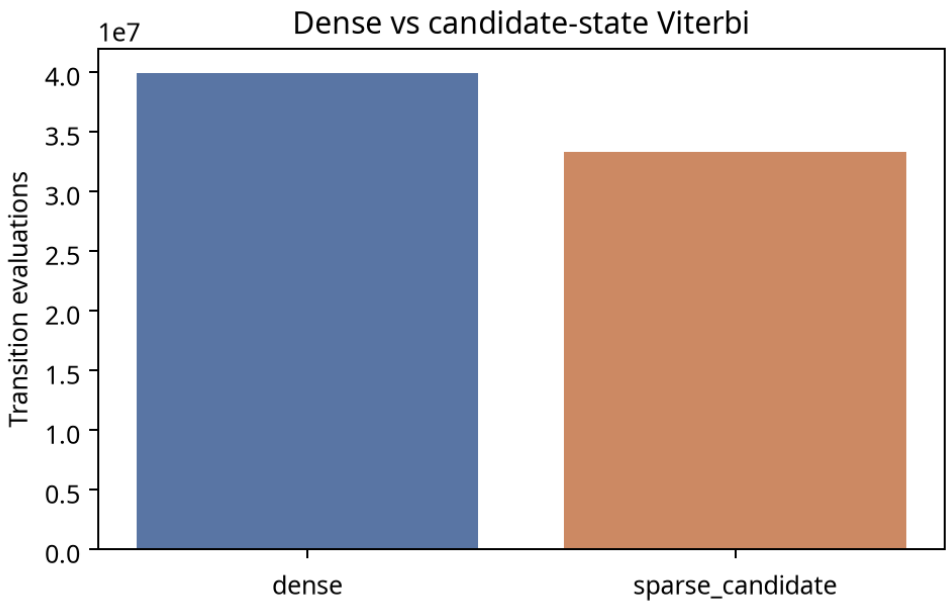
The design exploits lexical and suffix-conditioned emission sparsity. It stores candidate tag sets and adds a fallback path; it does not replace the HMM probability model. Trade-offs: training support can omit a test-time gold tag for a known word; overly tight candidate sets reduce accuracy. Signature buckets and dense fallback control this risk.

7.D Prediction before implementation

Prediction: sparse candidate-state Viterbi will evaluate fewer transitions per token than dense Viterbi, with unchanged accuracy on known words whose observed tag set includes the gold tag. Accuracy may fall slightly on ambiguous and unseen tokens if candidate sets omit a valid tag. The experiment must count transitions and report accuracy on the identical test split.

7.E Implementation and validation

Decoder	Tokens	Accuracy	Transition evaluations	Evaluations/token
dense	19655	0.93788	39969170	2033.54
sparse_candidate	19655	0.93788	33294142	1693.93



Counted reduction: 16.70% fewer transitions under the candidate decoder on the same test sample; accuracy is 0.93788 for dense and 0.93788 for sparse decoding. The 50% reduction objective was not achieved, although the measured accuracy was unchanged. Candidate sets were still broad for ambiguous words; this redesign is safe here but not a sufficient optimization. A next experiment should combine candidate pruning with entropy/confidence thresholds and retain the dense fallback.

7.F Defence and trade-offs

When lexical candidate coverage is high, sparse decoding avoids scores for impossible word-tag emissions and retains HMM transition ranking among plausible tags. The dense decoder remains preferable for tiny tagsets, very ambiguous vocabularies, or when no reliable candidate set exists. The redesign is not “better” merely because it is sparse: the report compares accuracy and transition counts together. It remains a local sequence model and does not solve long-distance syntax, code-mixed morphology or domain shift without additional features and evaluation.

Appendix A – Reproducible corpus and data manifest

Artifact	Source / generation	Size / split	Use
Brown reviews	Brown Corpus via NLTK [1] [2]	40,704 words in installed reader; sentence split 80/20 seed 42	Q1; matching LM condition
Brown news	Brown Corpus via NLTK [1] [2]	4,623 sentences; random 80/20	Q2/Q3/Q4 out-of-domain
Synthetic tickets	15 fixed support templates, slot variation, seed 42	1,800 sentences; random 80/20	Q2 in-domain controlled proxy
Repeated edge	Two near-identical phrases repeated 500 times each	1,000 sentences; random 80/20	Q2 sparsity stress test
Penn Treebank	NLTK treebank tagged sentences [2][3]	3,914 sentences; 100,676 tokens; random 80/20	Q5/Q6 tagging
Chat annotation worksheet	100 generated support-chat sentences	Initial labels only; second annotator blank	Q5 annotation handoff

The NLTK corpus readers support raw text, word, sentence and tagged-sentence access, including Brown tagged words and Penn Treebank. [2] Corpus versions and category filters should be recorded when reproducing the run. Brown is edited 1961 American English, not modern telecom support; it is a public proxy rather than a domain match. [1]

A.1 Data leakage controls

- Split at sentence/document level before n-gram and tagger training.
- Fit vocabularies, word frequency cutoffs, tag lexicons and interpolation weights only from training data.
- Keep Brown reviews and news conditions separate; do not tune against the reported news test set.
- Record random seed, package versions, corpus counts and script hash.
- For telecom evaluation, replace synthetic tickets with de-identified, consented support conversations and group split by customer/ticket.

Appendix B – Five example tokenizations and expanded decisions

Can't connect!! Net speed 4G/5G isn't consistent :(
N=12; V=11; tokens: can't | connect | ! | ! | net | speed | 4g | / | 5g | isn't | consistent | :(

Won't recharge via e-wallet
N=4; V=4; tokens: won't | recharge | via | e-wallet

Rs.499 plan is value-for-money
N=4; V=4; tokens: rs.499 | plan | is | value-for-money

It is the-best-network, no? ok ok
N=8; V=7; tokens: it | is | the-best-network | , | no | ? | ok | ok

Super service da, semma speed
N=6; V=6; tokens: super | service | da | , | semma | speed

The plan is not bad-fast?
N=6; V=6; tokens: the | plan | is | not | bad-fast | ?

B.1 Ambiguous string analysis

“I saw her duck :)” is tokenized deterministically as i / saw / her / duck / :). The tokenizer cannot decide whether saw is a verb or noun or whether duck is a noun or verb. “Her” also alternates between determiner and pronoun uses. A POS tagger resolves the sequence statistically or by rules; the ambiguity is linguistic, not a token-counting bug.

“Rs. 499/5G plan” produces a currency number, slash, 5G and plan under this policy. Whether the slash is a fraction marker, a separator or part of a model name requires context. Keep original text and record normalization decisions to avoid irreversible segmentation.

Appendix C – Full n-gram raw results

synthetic helpdesk

n	Distinct	Test events	Unseen	Unseen %	Zero sentences	Greedy sample
1	94	3543	0	0.00%	0	. . . . .
2	159	3543	0	0.00%	0	the app says payment pending after recharge .
3	203	3543	0	0.00%	0	the app says payment pending after recharge .
4	232	3543	0	0.00%	0	the app says payment pending after recharge .

Brown news OOD

n	Distinct	Test events	Unseen	Unseen %	Zero sentences	Greedy sample
1	11420	21548	1435	6.66%	634	the the the the the the the the the the
2	50912	21548	10276	47.69%	914	the first time , and the first time , and
3	74651	21548	17064	79.19%	918	the president and voting to cut off legislation- delaying filibusters ?
4	81356	21548	19361	89.85%	918	the president said he will recommend later an increase in

repeated phrase edge

n	Distinct	Test events	Unseen	Unseen %	Zero sentences	Greedy sample
1	9	1800	0	0.00%	0	your your your your your your your your your
2	13	1800	0	0.00%	0	reset your router and restart the modem .
3	16	1800	0	0.00%	0	reset your router and restart the modem .
4	18	1800	0	0.00%	0	reset your router and restart the modem .

The sample generator uses the highest-count continuation within the longest available context; it is a diagnostic illustration, not a good text-generation algorithm. The edge case is deliberately duplicated and should not be interpreted as generalization performance.

Appendix D – Smoothing probability and entropy details

D.1 Conditional probability checks

add1	history	sum over mapped vocabulary
add1	<UNK>	1.0000000000
add1	the	1.0000000000
add1	,	1.0000000000
add1	.	1.0000000000
add1	<s>	1.0000000000
add1	of	1.0000000000
add1	and	1.0000000000
add1	a	1.0000000000
add1	'	1.0000000000
add1	`	1.0000000000

addk	history	sum over mapped vocabulary
addk	<UNK>	1.0000000000
addk	the	1.0000000000
addk	,	1.0000000000
addk	.	1.0000000000
addk	<s>	1.0000000000
addk	of	1.0000000000
addk	and	1.0000000000
addk	a	1.0000000000
addk	'	1.0000000000
addk	`	1.0000000000

backoff	history	sum over mapped vocabulary
backoff	<UNK>	1.0000000000
backoff	the	1.0000000000
backoff	,	1.0000000000
backoff	.	1.0000000000
backoff	<s>	1.0000000000
backoff	of	1.0000000000
backoff	and	1.0000000000
backoff	a	1.0000000000
backoff	'	1.0000000000
backoff	`	1.0000000000

deleted	history	sum over mapped vocabulary
deleted	<UNK>	1.0000000000
deleted	the	1.0000000000
deleted	,	1.0000000000
deleted	.	1.0000000000
deleted	<s>	1.0000000000
deleted	of	1.0000000000
deleted	and	1.0000000000
deleted	a	1.0000000000
deleted	'	1.0000000000
deleted	`	1.0000000000

D.2 Toy entropy arithmetic

uniform10: H=3.3219281 bits; 2^H=10.0000000.

skewed_one_0.7: H=1.8322684 bits; 2^H=3.5609654.

For a uniform distribution over ten outcomes, every outcome has probability 0.1 and $H=-10\times0.1\times\log_2(0.1)=\log_2 10=3.321928$ bits, giving PP=10. For a skewed distribution with one outcome at 0.7 and nine at 0.3/9, entropy and perplexity are lower because uncertainty is concentrated on fewer effective outcomes.

D.3 Smoothing model definitions and limitations

Add-one: $P1(w|h) = (C(h,w)+1)/(C(h)+V)$
Add-k: $Pk(w|h) = (C(h,w)+k)/(C(h)+kV)$
Absolute discount: $P(w|h) = \max(C(h,w)-D,0)/C(h) + \alpha(h)P_lower(w)$
Deleted interpolation: $P(w|h) = \Sigma j\ \lambda j\ Pj(w|suffix_j(h))$; $\Sigma j\ \lambda j=1$

Add-one's unseen mass increases with vocabulary size. A discounting method moves a count-based mass budget to lower orders, while deleted interpolation selects weights according to leave-one-out support. Standard smoothing comparisons show that performance depends on corpus, order and estimator; no single method wins every condition. [6] The NLTK language-model API documents MLE, Laplace/Lidstone, interpolated and discounting estimators. [9]

Appendix E – Tagset and annotation reference

Custom tag	Definition	Example	Extraction use
PROD	Provider, device, plan or product name	Airtel, MyJio, OnePlus	Product/entity retrieval
NET_ABBR	Network/support abbreviation or code	4G, LTE, SIM, OTP	Normalize domain shorthand
COMPLAINT_V	Verb/state indicating failure or complaint	drops, failed, disconnect	Complaint/action extraction
EMOT	Emoticon or emoji	:(, 😂	Sentiment signal
CODEMIX	Romanised regional-language token	da, semma, ku	Language-aware routing
NUMCURRENCY	Price/amount expression	Rs.499	Price extraction

Universal POS is defined with open-class, closed-class and “other” tags; the Universal Dependencies list includes ADJ, ADV, NOUN, PROPN and VERB as open classes and ADP, AUX, CCONJ, DET, PRON and others as closed classes. [5] The assignment’s custom tagset is deliberately task-oriented and should be mapped to a stable base tagset when used outside the telecom domain.

E.1 Annotation worksheet status

The supplementary CSV `q5_100_chat_annotations.csv` has 100 synthetic support-chat sentences and initial rule/lexicon labels. The `independent_annotator_B` column is intentionally blank. It must be filled by a peer or a later independent pass before calculating Cohen's κ. Agreement statistics are not inferred from deterministic tagset mappings, because that would not represent independent human annotation.

Appendix F – Extended POS-tagging results

10% training data

Tagger	Acc	Known	Unknown	Known n	OOV n	Rules	Transitions	Top errors
MFS	0.77125	0.93281	0.21114	15255	4400	0	0	NNP>NN:1034;JJ>NN:547;NNS>NN:522;CD>NN:312;VBN>NN:231
Rule_based	0.80549	0.85834	0.62227	15255	4400	0	0	JJ>NN:416;VBN>VBD:354;-NONE->CD:190;-NONE->.:190;VBZ>NNS:182
Bigram_HMM_Viterbi	0.87744	0.93873	0.74961	13285	6370	0	33323094	NN>JJ:219;JJ>NN:150;NN>NNP:87;NNP>JJ:62;VBN>VBD:59
Brill_TBL	0.78433	0.94081	0.24182	15255	4400	17	0	NNP>NN:1025;JJ>NN:546;NNS>NN:516;CD>NN:293;VBN>NN:231

50% training data

Tagger	Acc	Known	Unknown	Known n	OOV n	Rules	Transitions	Top errors
MFS	0.86121	0.93991	0.18497	17606	2049	0	0	NNP>NN:473;JJ>NN:328;NNS>NN:249;CD>NN:191;VB>NN:160
Rule_based	0.82564	0.85158	0.60273	17606	2049	0	0	VBN>VBD:356;JJ>NN:236;-NONE->CD:190;-NONE->.:190;VBZ>NNS:182
Bigram_HMM_Viterbi	0.92699	0.94978	0.80659	16527	3128	0	38251035	NN>JJ:102;NN>NNP:87;JJ>NN:82;NNP>NN:74;JJ>NNP:36
Brill_TBL	0.88074	0.95723	0.22352	17606	2049	40	0	NNP>NN:470;JJ>NN:327;NNS>NN:247;CD>NN:181;VBG>NN:117

100% training data

Tagger	Acc	Known	Unknown	Known n	OOV n	Rules	Transitions	Top errors
MFS	0.89031	0.94384	0.17471	18287	1368	0	0	NNP>NN:307;JJ>NN:257;NNS>NN:168;CD>NN:150;VB>NN:134
Rule_based	0.83495	0.85339	0.58845	18287	1368	0	0	VBN>VBD:356;-NONE->CD:190;-NONE->.:190;VBZ>NNS:182;JJ>NN:178
Bigram_HMM_Viterbi	0.93788	0.95376	0.80908	17497	2158	0	39969170	NN>NNP:87;NNP>NN:86;NN>JJ:82;JJ>NN:62;VBD>VBN:39
Brill_TBL	0.90867	0.96068	0.21345	18287	1368	40	0	NNP>NN:304;JJ>NN:251;NNS>NN:167;CD>NN:141;VBG>NN:90

Accuracy is token-level exact tag accuracy. Known/OOV status is defined against the training vocabulary. The rule tagger uses a lexicon plus ordered handcrafted rules; its OOV count is measured relative to that training lexicon. The Brill rule cap is fixed at 40. Brown/Treebank tag conventions are not interchangeable without mapping; the Penn Treebank sample is retained in its native tagset for this experiment. [2] [3]